Note

- The Highest Common Factor (HCF) is also called as the Greatest Common Divisor (GCD) or the Greatest Common Factor (GCF).
- HCF (1, x) =1
- HCF (x, y) = x, if y is a multiple of x. For example, HCF (3, 6) = 3.
- If the HCF of two numbers is 1, then the numbers are said to be **co-primes** or **relatively prime.** Here, the two numbers can both be primes as (5, 7) or both can be composites as (14, 27) or one can be a prime and other a composite as (11, 12).

Example 3: Find the HCF of the numbers 40 and 56 by division method.

Solution:

2	40
2	20
2	10
5	5
	1

2	56
2	28
2	14
7	7
	1

or

2	40, 56
2	20, 28
2	10, 14
	5, 7

Prime factorisation of 40 = 2 x 2 x 2 x 5
Prime factorisation of 56 = 2 x 2 x 2 x 7
The product of common factors of 40 and 56
= 2 x 2 x 2 = 8 and so, HCF (40, 56) = 8

Dividing by the common factor 2, (in 3 steps)
HCF = Product of common factors
= 2 x 2 x 2 = 8

Example 4: Find the HCF of the numbers 18, 24 and 30 by factor tree method.

Solution:

Let us find the factors of 18, 24 and 30 (use of divisibility test rules will also help).

The factors of 18 are 1, 2, 3, 6, 9 and 18.

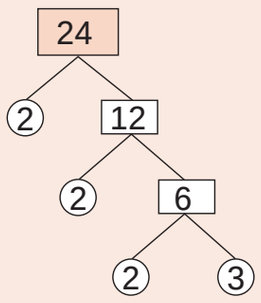
The factors of 24 are 1, 2, 3, 4, 6, 8, 12 and 24.

The factors of 30 are 1, 2, 3, 5, 6, 10, 15 and 30.

The factors that are common to all the three given numbers are 1, 2, 3 and 6 of which 6 is the highest. Hence, HCF (18, 24, 30) = 6.

Note that 1 is a trivial factor of all numbers.

Let us find the factors of 24 by tree method.



Here, 24 = 2 x 2 x 2 x 3
Similarly, we can find the factors of 18 and 30.

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